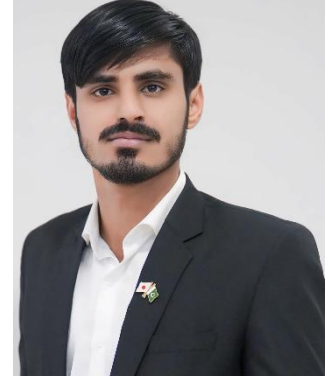

CURRICULUM VITAE – Muhammad Nazim

PERSONAL Information

Muhammad Nazim
Nationality: Pakistani.
Sex: Male.
CNIC: 32304-5565383-7.
Born: 10/08/2000, in Punjab province,
Muzaffargarh 34000, Pakistan.
Phone number: +923080743428, WhatsApp
account is available.
Email: muhammad.nazim.researcher@gmail.com.
Current Position: Researcher at IIUI.
Current address: International Islamic
University Islamabad, sector H-10, Islamabad
Pakistan.



Professional Objective

A strong aspiration exists to explore daily scientific advancements that benefit society. This includes developing effective methodologies to address situational challenges, sharing mathematical knowledge globally, and generating sustainable revenue to support the Society's initiatives. It also involves promoting mathematical research and demonstrating its value to decision-makers, policymakers, funders, and users of mathematics. The overarching objective is to advance mathematical understanding by fostering research, collaboration, and support for mathematicians. Additionally, there is a deep interest in discovering new insights through the application of partial differential equations and examining their use across diverse fields to enhance mathematical contributions in multiple domains.

Education

Master in Mathematics: Jan 2026, International Islamic University Islamabad, Pakistan.
Bachelor Degree in applied mathematics: Jan 2023, International Islamic University Islamabad, Pakistan.
Fsc: July 2018, DG Khan board of intermediate and secondary education.
Metric: July 2016, DG Khan board of intermediate and secondary education.

Skills

Scientific software: Tensor Flow, Mathematica, MATLAB, PyTorch
Programming languages: Python using google colab.
Physics Informed Neural Network.
Machine learning.
Deep Learning.
Artificial Neural Network.
Numerical Analysis.
Office software: Microsoft Office.
Language: Urdu, English.

Publications

1. **Thesis Title:** (Master thesis title) Analysis of Parameter Uncertainty on Stochastic Heat Transfer Problem.
2. **Muhammad Nazim**, Muhammad Imran Khan, Ahmed Zeeshan, ‘Hybrid Stochastic Deterministic Framework for Uncertainty Propagation in Dynamic Relaxation Systems Using Monte Carlo and Polynomial Chaos Expansion’ (Revision is submitted Nonlinear science).
3. **Muhammad Nazim**, Mohammed El Khider, Mohammed, Almakki, Muhammad Imran Khan, Muhammad Shoaib, ‘Stochastic Uncertainty Alignment in Physics-Informed Neural Network: An Optimization of Two-Phase Dusty Flow’ (Under view at advance theory and simulation).
4. Muhammad Shaoaib, Mohammed, Almakki, Mohammed El Khider, Muhammad Imran Khan, **Muhammad Nazim**, ‘Physics-Informed Neural Network for Stochastic Realization of Mixed Convection Flow: An ADAM Optimization Perspective of Partial Differential Equation’ (submitted at international journal of mathematics and computer science).
5. Muhammad Imran Khan, **Muhammad Nazim**, and Ahmed Zeeshan, ‘Stochastic Alignment in Heat and Fluid Flow over a Circular Cylinder with Physics Informed Neural Network: A second Order Optimization Perspective’ (Under review at engineering applications of artificial intelligence). (Corresponding author: **Muhammad Nazim**).
6. **Muhammad Nazim**, Amad Ur Rehman, Ahmed Zeeshan, H.Fish, S.U.Khan and M.Helal, ‘Hybrid Stochastic Modeling of Heat Transfer Using Non-Intrusive Spectral Projection and Artificial Neural Networks for Engineering Applications’ (submitted at case studies in Thermal Engineering).(Corresponding authoer: Dr.Fateh Mebarek Oudina).
7. Amad Ur Rehman, **Muhammad Nazim**, Ahmed Zeeshan and Nasir Shehzad, ‘A Physics-Informed Neural Network Study of MHD non-Newtonian Nanofluid Flow with Heat and Mass Transfer in a Sinusoidally Varying Channel’ (Complete research monograph / technical book draft), 202X.

Recent New Publications & Submissions (2026)

Muhammad Nazim, ‘Physics Informed Deep Learning Based Stochastic Optimization for Thermal Performance of Nonlinear Two-Phase Dusty Nanofluid Flow Dynamics in a Porous Media’ (Under review at Journal of Porous Media; Handling Editor: Dr. Dia Zeidan). [TOP PRIORITY]

Muhammad Nazim, ‘Bayesian Stochastic Physics-Informed Neural Networks with Adaptive Residual Learning for Uncertainty Quantification of Intravitreal Drug Transport’ (Submitted at Applied Soft Computing Journal).

Muhammad Nazim, ‘Uncertainty Quantification with Physics-Informed Learning for Diagnosis of Coronary Artery Stenosis’ (Submitted at Biomedical Signal Processing and Control).

Muhammad Nazim, ‘Nonlinear Ciliary Flagellar Flow Dynamics and Energy Transport in a Curved Porous Channel: An Optimization via Physics Informed Neural Network’ (Published).

Muhammad Nazim, Amad Ur Rehman, Ahmed Zeeshan, H.Fish, S.U.Khan and M.Helal, ‘Uncertainty Quantification in Heat Transfer: A Stochastic Approach to Fins and Forced Convection Over Cylinders’ (Submitted at Case Studies in Thermal Engineering).

Muhammad Imran Khan, Muhammad Nazim, ‘Non-Intrusive Polynomial Chaos Expansion for Stochastic Simulation of Bioheat Equation’ (Under review).

Muhammad Nazim, ‘Parametric Stochastic Physics-Informed Neural Networks for Uncertainty Quantification of Thermo-Fluid Flow over Cylindrical Surfaces’ (Under review).

Muhammad Nazim, ‘Stochastic Physics-Informed Neural Networks for Uncertainty-Aware Prediction of Thermo-Fluid Systems’ (Under review).

***SciEntific
InterESTs***

- Simulation of Stochastic problem.
 - Machine learning techniques to solve the nonlinear partial differential equations.
 - Artificial neural networks-based simulation of boundary value problems
 - Mathematical Analysis, Applied Mathematics.
 - Numerical Simulation of Boundary layer flow.
 - Finite difference and finite element method.
 - Stability analysis with eigenvalue problem.
 - Sensitivity analysis of boundary layer problem.
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ReFeRENCES

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